Damped two-stream couplings in equilibrium reasoners: attractor geometry, inner loops and test-time scaling on Sudoku-Extreme

Håkon Noren

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Abstract

Equilibrium Reasoners (EqR) iterate one shared block on two latent states and reach high accuracy on Sudoku-Extreme by spending test-time compute: more iterations (depth), more random starts (breadth), and Langevin noise. We replace EqR's self-reading update by two-stream *kick* and *relax* couplings in which each stream is driven only by the other, so that every update is an invertible shear followed by a constant scaling, and we study which properties of the resulting maps decide accuracy, depth scaling, breadth and the effect of noise. On the theory side we show that (i) the force block has unit per-token RMS, so the state at a fixed point is an exact, rate-determined multiple of a unit vector, which fixes the scale against which test noise acts; (ii) the volume constraint of the coupled maps is not binding for any trained model except one with near-identity rates; (iii) symmetric relaxation without common-mode damping has an exactly neutral mode, and damping it is what makes relax work; (iv) alternating couplings have no implicit inner fixed-point problem, while EqR's inner loop does; and (v) a relaxed self-map with strong negative feedback undergoes a period-doubling (flip) bifurcation, which a two-copy ("twin") coupling turns back into a fixed point. Numerically, on 14 checkpoints we measure Jacobian spectra, path independence, selection, and escape under noise. Right answers are attracting fixed points and wrong end states are not, in every working model, so convergence-based selection is nearly perfect and accuracy is limited by *coverage*. Test noise helps by selectively destabilising wrong attractors below an arm-specific breakdown level. EqR differs from the coupled maps in being near critical at its answers ($\rho \approx 0.93$–0.97 per outer iteration against 0.53–0.73), in being more path independent, and in an inner loop that, with training, goes through a flip bifurcation into an exact 2-cycle. With training to 50k steps, EqR pulls ahead of the coupled maps on every metric. All results are single-seed.

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1 Introduction

Looped (weight-tied, recurrent-depth) reasoners solve constraint problems such as Sudoku by iterating one block many times [8, 12, 13, 18]. Equilibrium Reasoners (EqR, [12]) add three ingredients to TRM-style recursion: random initial states, Langevin noise in the state, and an explicit view of reasoning as convergence to an attractor. At test time EqR scales *depth* (iterations), *breadth* (random starts per puzzle, with an answer chosen by convergence) and noise.

This note asks which properties of the iteration map make these work. We keep EqR's data, block, training recipe and evaluator, and replace only the state update by two-stream couplings in the spirit of conformal symplectic integrators [7, 15] and momentum networks [17]:

$$\begin{aligned} \textbf{kick:}\quad & L \leftarrow a\,(L + \mathcal{B}(H + x)), && H \leftarrow bH + \mathcal{B}(L + x), \\ \textbf{relax:}\quad & L \leftarrow aL + (1-a)(cH + \mathcal{B}(H + x)), && H \leftarrow bH + (1-b)(cL + \mathcal{B}(L + x)), \end{aligned}$$

with $a = e^{-\gamma}$, $b = e^{-\alpha}$, $c = e^{-\delta}$. Each stream is driven only by the other stream, so every update is invertible with a state-independent Jacobian determinant. We then compare these maps with EqR in accuracy (§4), test-time scaling (§5) and a direct dynamical analysis of trained checkpoints (§6), guided by a small amount of theory (§3).

Summary of findings.

1. **Certification is easy; coverage is hard.** In every working model right answers are attracting fixed points ($\rho < 1$) and wrong end states are not ($\rho > 1$). Convergence, the halting head and a Sudoku validity check therefore pick the right start almost whenever one exists (AUROC ≈ 1); what separates models is how often random starts land in the right basin (§6.1).
2. **Noise acts on wrong attractors first.** Under noise σ below an arm-specific breakdown level $\sigma^\star$ right answers are never lost, while 15–33% of wrong end states are rescued (§6.3), as in Hopfield networks at finite temperature [1, 6]. $\sigma^\star$ depends on the state scale, which is fixed exactly by the rates (Prop. 2), and on how strongly the answers attract.
3. **EqR is near critical and more path independent.** EqR's right answers have $\rho \approx 0.87 \to 0.93 \to 0.97$ (20k, 50k, released) per outer iteration, matching the 0.89–0.97 reported for converged Sudoku loopers [14]; the coupled maps contract much harder (0.53–0.73). EqR's

random starts reach fewer distinct answers (1.6 per puzzle at 50k against 2.6 for relax and 3.2 for kick).

4. **EqR's inner loop flips.** With z_H frozen, EqR's inner map has a stable fixed point at 20k steps but sits on an exact 2-cycle at 50k (Prop. 8); its leading Jacobian eigenvalue moves from $\approx \pm 0.6$ to ≈ -1 and, in the released checkpoint, to ≈ -2. The cycle is invisible to the readout: odd inner-loop lengths work as well as even ones.
5. **Damping matters, volume does not.** The volume bound of the coupled maps (Prop. 4) is irrelevant for all trained models except a near-identity kick; relax without common-mode damping has an exactly neutral mode (Prop. 5) and fails, and accuracy improves monotonically as the merged-state retention falls from 1 to ≈ 0.2 (§4).
6. **Training length favours EqR.** At 20k steps relax ties EqR with enough test-time compute; at 50k EqR leads on every metric (depth 1024: 97.1 vs. 92.5; depth 64 with 128 starts: 98.7 vs. 96.6).

2 Setting

2.1 EqR

A puzzle is embedded as $x \in \mathbb{R}^{n \times d}$ ($n = 97$ positions: 16 puzzle-embedding slots and 81 cells; $d = 512$). The shared block $\mathcal{B} : \mathbb{R}^{n \times d} \to \mathbb{R}^{n \times d}$ is two post-norm layers, each $v \mapsto N(v + \mathrm{MLP}_t(v))$ followed by $v \mapsto N(v + \mathrm{SwiGLU}(v))$, where MLP_t mixes tokens and N is RMS normalisation *without gain*, $N(y)_i = y_i / \operatorname{rms}(y_i)$ row-wise.

Definition 1 (EqR update [12])**.** With $\lambda = 0.95$ and noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$ drawn afresh after every update, one *latent recursion* is

$$z_L \leftarrow (1-\lambda) z_L + \lambda\, \mathcal{B}(z_L + z_H + x) + \xi \quad \text{(6 times)}, \qquad z_H \leftarrow (1-\lambda) z_H + \lambda\, \mathcal{B}(z_H + z_L) + \xi.$$

One *outer iteration* (one ACT step) is three recursions, i.e. 21 block evaluations; the readout decodes z_H. Training differentiates only the last recursion (7 evaluations), carries the state across outer iterations detached, starts from random states (truncated normal, std 1), and uses $\sigma = 0.01$.

2.2 Coupled updates

Definition 2 (Kick and relax)**.** Let $a = e^{-\gamma},\ b = e^{-\alpha},\ c = e^{-\delta} \in (0, 1]$. A *half-step* updates one stream from the other:

$$\begin{aligned} \text{kick:} \quad & L \leftarrow a\,(L + \mathcal{B}(H + x)), && H \leftarrow bH + \mathcal{B}(L + x), \\ \text{relax:} \quad & L \leftarrow aL + (1-a)(cH + \mathcal{B}(H + x)), && H \leftarrow bH + (1-b)(cL + \mathcal{B}(L + x)). \end{aligned}$$

A recursion is the alternating sequence H, L, H, L, H, L, H (7 evaluations); three recursions form an outer iteration (21 evaluations, 7 differentiated), exactly as in EqR. The readout decodes $N(H)$. Noise is added after every half-step.

Variants (Table 1): $x \to L$ *only* drops x from H's force (EqR's choice); *self-read* adds the updated stream to its own force input ($L \leftarrow \ldots \mathcal{B}(L + H + x)$), which breaks invertibility; *inner loop* uses EqR's order ($L^6 H$) with self-read; *twin* doubles L into L, L' that read each other (§3.4).

Table 1: Arms. Rates are fixed unless stated. "Positive" kick learns $\gamma = 0.05 + \text{softplus}(\cdot)$ from $\gamma = 1$. Invertible: every update is a shear followed by a constant scaling.

arm	update	rates	invertible	round
EqR	self-read, order L^6H	$1 - \lambda = 0.05$	no	ref.
kick: fixed, learned, positive	kick	γ=α=0.005; learned; learned ≥ 0.05	yes	1
relax, $\delta \in \{0, \frac{1}{4}, 1\}$	relax	$\gamma = \alpha = 3$	yes	1
relax / kick, $x \to L$ only	no x in H's force	as above	yes	2
relax δ=1, $x \to L$, self-read	forces read own stream	$\gamma = \alpha = 3$	no	2
relax, $\delta \in \{2, 3\}$	relax	$\gamma = \alpha = 3$	yes	3
twin L, $\delta \in \{1, 2\}$	L, L' read each other, order $LL'LL'LL'H$	$\gamma = \alpha = 3$	yes*	3
inner loop, self-read	self-read relax, order L^6H	$\gamma = \alpha = 3$, $\delta = 1$	no	3

*Every update reads only streams it does not change; no inverse is implemented.

2.3 Evaluation

Depth D is the number of outer iterations; *breadth* B is the number of random starts per puzzle. For $B > 1$ an answer is chosen by: the most converged start (EqR's score, the logit change over the last three iterations; "converged"); the mean accuracy of the four most converged ("top-4", EqR's headline); plurality of full grids ("vote"); the largest halting logit; or a label-free *Sudoku verifier* (a grid with no rule violations that keeps the givens; Sudoku-Extreme puzzles have unique solutions, so this equals "some start is right", "any"). Test noise defaults to $\sigma = 0.5$ as in EqR's released evaluation. Sample sizes: whole test set (423k) for in-training accuracy at 16 iterations, the first 2048 puzzles for depth×breadth, and the first 256 puzzles ×16 starts ×256 iterations for the dynamical analysis.

3 Theory

3.1 State scale

Lemma 1 (Unit-RMS force)**.** *Every token row of $\mathcal{B}(u)$ has RMS exactly one (up to the normalisation ϵ).*

Proof. $\mathcal{B}$ ends with N, which divides each row by its RMS and has no gain parameter. □

Proposition 2 (Fixed-point scale)**.** *At a noise-free fixed point, with f the (unit-RMS) force evaluated there:*

(a) EqR: $z_L^* = \mathcal{B}(\cdot)$, so $\text{rms}(z_L^*) = 1$, and likewise z_H^*;

(b) kick: $L^* = \frac{a}{1-a} f_L$ and $H^* = \frac{1}{1-b} f_H$, so $\text{rms}(L^*) = \frac{a}{1-a}$, $\text{rms}(H^*) = \frac{1}{1-b}$;

(c) relax with merged streams $H^* = L^* = z^*$: $z^* = \frac{1}{1-c}\mathcal{B}(z^* + x)$, so $\text{rms}(z^*) = \frac{1}{1-c}$; for $c = 1$ there is no fixed point unless $\mathcal{B}(z^* + x) = 0$.

Proof. Set new state equal to old state in each update and solve; e.g. for kick $L^* = a(L^* + f_L)$ gives $L^* = \frac{a}{1-a} f_L$; for merged relax $z^* = az^* + (1-a)(cz^* + f)$ gives $(1-a)(1-c)z^* = (1-a)f$. Lemma 1 gives the RMS. □

The identity is exact per token. The measured endpoint RMS matches it to within 1% for every EqR, kick and merged-relax checkpoint (Figure 3, Table 9; it does not apply to the unmerged $x \to L$ relax arms or to $\delta = 0$, which has no fixed point): e.g. kick (positive, 20k) L: predicted

0.64, measured 0.64; merged relax $\delta = 1$: 1.58 vs. 1.58; relax $\delta = \frac{1}{4}$: 4.52 vs. 4.51; kick (fixed): 200 vs. 199.8.

Corollary 3 (Relative noise)**.** *Isotropic noise σ acts on a stream of scale s as relative noise σ/s. At $\sigma = 0.5$ this is 0.5 for EqR, ≈ 0.3 for merged relax ($\delta = 1$), and ≈ 0.8 for the L stream of positive kick at 20k (whose L grows to 0.81 by 50k as γ falls).*

3.2 Volume

Proposition 4 (Volume bound)**.** *Each coupled half-step is a shear $(s, o) \mapsto (s+g(o), o)$ composed with a constant diagonal scaling, so for one outer iteration with n_L L-half-steps and n_H H-half-steps (per channel and token) $\log|\det J| = -nd\,(n_L\gamma + n_H\alpha)$. Hence the geometric mean of $|\mu|$ over the $2nd$ eigenvalues of J is $e^{-(n_L\gamma+n_H\alpha)/2}$, and in particular $\rho(J) \geq e^{-(n_L\gamma+n_H\alpha)/2}$ at every fixed point.*

Proof. A shear has unit Jacobian determinant; the scaling contributes a^{nd} or b^{nd} per half-step (for relax the cross term $(1-a)cH$ is part of the shear). Multiply over the 21 half-steps. The spectral radius is at least the geometric mean of the moduli. □

Remark 1. The shears are symplectic only if $\partial g/\partial o$ is symmetric, which a learned block is not, so the maps are volume-contracting at a constant rate but not conformal symplectic: there is no eigenvalue pairing $\mu \leftrightarrow e^{-\cdot}/\mu$ [15], only the determinant constraint.

The bound is 2×10^{-14} per outer iteration for relax (rate 3), 9×10^{-6} and 7×10^{-7} for positive kick (20k, 50k), 0.017 for learned kick, but 0.949 for fixed kick ($\gamma = \alpha = 0.005$). Only the last is constrained: it cannot contract, and indeed its end states have $\rho = 1.01$ and it keeps a right answer only 42% of the time even at $\sigma = 0.1$ (Table 7).

3.3 The neutral mode of undamped relaxation

Proposition 5 (Common mode)**.** *With the forces frozen, the linear part of the relax half-steps acting on (L, H) per channel is $M_L = \left(\begin{smallmatrix} a & (1-a)c \\ 0 & 1 \end{smallmatrix}\right)$, $M_H = \left(\begin{smallmatrix} 1 & 0 \\ (1-b)c & b \end{smallmatrix}\right)$. For $c = 1$ both are row-stochastic, so $(1, 1)^\top$ is a fixed vector of every product: the common mode is exactly neutral ($\mu = 1$). For $c < 1$ the row sums are $a + (1-a)c < 1$ and $b + (1-b)c < 1$, so every product has spectral radius below one. With $a = b$ and merged streams the common mode is retained at the rate $r = a + (1-a)c$ per half-step.*

Proof. Row-stochastic matrices have eigenvalue 1 with eigenvector $\mathbf{1}$; a nonnegative matrix's spectral radius is at most its maximum row sum. □

Relax with $\delta = 0$ therefore drifts along the force (endpoint RMS $\approx 4 \times 10^3$, $\rho = 1.00$ at its end states) and reaches only 13%. Accuracy rises as the retention r falls (Figure 2); EqR's own self-retention is $1 - \lambda = 0.05$.

3.4 Inner fixed-point problems

Proposition 6 (Alternating couplings have an explicit inner equilibrium)**.** *With H frozen, repeated L half-steps converge geometrically at rate a to $\Lambda(H) = cH + \mathcal{B}(H + x)$ (relax) or $\Lambda(H) = \frac{a}{1-a}\mathcal{B}(H + x)$ (kick), each a single block evaluation. In EqR the inner equilibrium solves the implicit equation $z_L = \mathcal{B}(z_L + z_H + x)$, which takes iteration.*

Proof. The force on L does not depend on L, so the L update is affine in L with slope a. □

In the language of two-timescale iterations [4], EqR's fast variable tracks an implicitly defined equilibrium $\Lambda(z_H)$ between slow updates, whereas in the alternating couplings Λ is one explicit layer, so the coupled models are effectively single-latent maps (and relax's streams merge, $\cos(H, L) = 1.00$). EqR's and TRM's ablations put the cost of a single latent at ≈ 10 points [12, 13]. The twin coupling restores an implicit inner problem:

Proposition 7 (Twins are a two-copy doubling)**.** *Let $G(z) = cz + \mathcal{B}(z + H + x)$ for frozen H. The twin updates $L \leftarrow aL + (1-a)G(L')$, $L' \leftarrow aL' + (1-a)G(L)$ with $a < 1$ have a fixed point (p, q) if and only if $p = G(q)$ and $q = G(p)$, i.e. p is a fixed point of $G \circ G$. Thus the twins represent both fixed points ($p = q$) and period-2 orbits ($p \neq q$) of the self-reading map G as fixed points.*

Proof. At a fixed point $p = ap + (1-a)G(q)$, i.e. $p = G(q)$, and symmetrically $q = G(p)$. □

Remark 2. This is the construction used to analyse synchronous Hopfield dynamics: a synchronously updated symmetric network converges to fixed points or 2-cycles, and its two-copy bipartite version, updated sequentially, has exactly these as fixed points [5, 11].

3.5 Flip bifurcation of relaxed self-maps

Proposition 8 (Period doubling)**.** *Let $F(z) = (1-\lambda)z + \lambda\tilde{\mathcal{B}}(z)$ and let μ be a real eigenvalue of $D\tilde{\mathcal{B}}$ at a fixed point z^*. Then $1 - \lambda + \lambda\mu$ is an eigenvalue of $DF(z^*)$; the direction is stable iff $1 - \frac{2}{\lambda} < \mu < 1$. As μ decreases through $1 - \frac{2}{\lambda}$ ($= -1.105$ at $\lambda = 0.95$) the eigenvalue of DF crosses -1, and generically a period-2 orbit is born (flip bifurcation). Smaller step sizes $\lambda < 2/(1-\mu)$ restore stability.*

Proof. $DF = (1-\lambda)I + \lambda D\tilde{\mathcal{B}}$ shares eigenvectors with $D\tilde{\mathcal{B}}$. The flip-bifurcation statement is the standard normal-form result for a simple eigenvalue crossing -1. □

Strong negative feedback ($\mu < -1$) is natural for Sudoku, whose constraints are mutual exclusions. The stability region and the measured values for EqR are drawn in Figure 13; damping of this kind is also the mechanism behind the fixed-point reasoners of [16].

3.6 Certification and noise

EqR bounds the distance to a fixed point by the residual under local contraction ($\|z - z^*\| \leq \|R(z)\|/(1-L)$) and turns this into a correctness certificate when the output margin is large [12, App. A.1]. The certificate needs right answers to be contracting and wrong ones not to be; we measure exactly this in §6.2. For noise, along a normal eigen-direction with eigenvalue μ the linearised stationary variance is $\sigma^2/(1-|\mu|^2)$: near-critical directions amplify noise, and a basin is left once the fluctuations reach its boundary. In Hopfield networks spurious mixture states lose stability at a lower temperature than retrieval states [1, 6]; noise then selects the right attractors. Section 6.3 tests this.

4 Accuracy and training

Table 2 and Figure 1 give the accuracies. Our EqR reproduces the paper (85.8 at 16 iterations and 92.1 at 64 after 50k steps, against 86.4 and 93.0 [12]). At 20k steps the two best couplings, relax $\delta = 1$ and positive kick, trail EqR by about six points at 16 iterations; the gap does not close with training (7.5 points at 50k) and widens with depth (10.5 points at 256 iterations). Fixed small rates fail (Prop. 4, 5).

Table 2: Accuracy at training noise ($\sigma = 0.01$), one start. Exact@16: whole test set; @64/@256: first 49,152 puzzles. O: right at 16 but wrong at 256; Δy: prediction changed between 16 and 256.

arm	steps	ex@16	cell@16	ex@64	ex@256	$O_{16\to256}$	$\Delta y_{16\to256}$
EqR	20k	69.5	89.3	73.2	75.8	1.7	31.3
EqR	50k	85.8	94.7	92.1	95.0	0.4	14.8
relax, $\delta = 1$	20k	63.9	87.3	65.4	66.4	2.2	36.6
relax, $\delta = 1$	50k	78.3	91.7	81.8	84.5	1.3	23.5
kick, positive rates	20k	63.5	87.2	65.0	66.1	1.9	35.8
kick, positive rates	50k	78.2	91.7	81.7	83.9	1.4	23.2
relax, $\delta = \frac{1}{4}$	20k	46.7	82.3	47.8	48.6	3.8	32.7
kick, learned rates	20k	46.5	82.1	47.3	48.0	2.0	37.2
relax, $\delta = 0$	20k	13.2	71.0	12.8	3.5	80.3	97.3
kick, fixed rates	20k	13.5	72.6	14.7	6.2	61.3	94.6
relax, $\delta = 1$, $x \to L$ only	20k	51.7	84.0	52.2	52.7	2.7	38.6
relax, $\delta = 1$, $x \to L$, self-read	20k	61.7	86.7	63.5	64.9	1.7	35.9
kick, positive, $x \to L$ only	20k	56.2	85.1	56.7	57.8	1.9	42.0

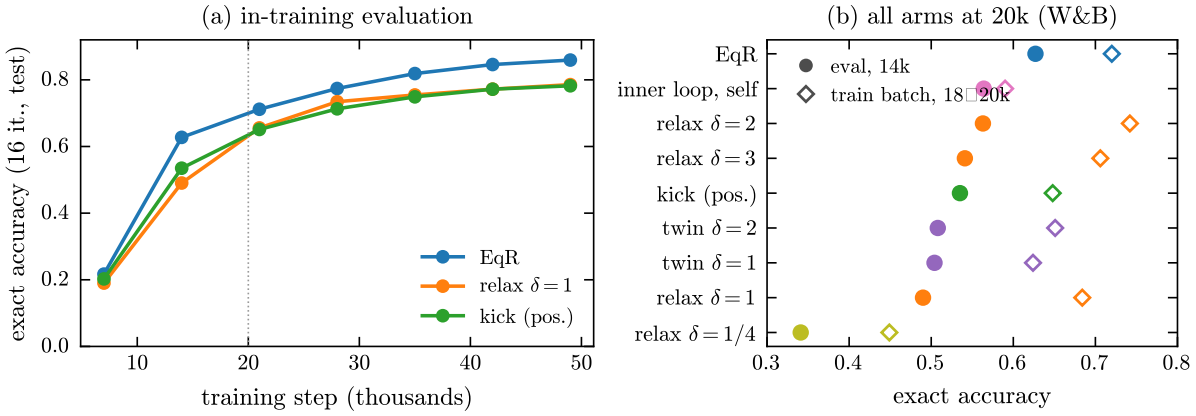


Figure 1: (a) In-training evaluation (whole test set, 16 iterations) of the three arms extended to 50k steps. (b) All round-three arms and references at 20k: evaluation at 14k (filled) and training-batch accuracy averaged over steps 18–20k (open).

Round two: input injection and self-reading. Giving x only to L (as EqR does) costs 12 points for relax and 7 for kick. For relax it separates the streams ($\cos(H, L)$ 1.00 $\to$ 0.77) but each stream is then driven only by the other. Adding self-reading recovers 10 of the 12 points (51.7 $\to$ 61.7), but only to the level of plain merged relax (63.9), which already reads itself in effect (its streams coincide). Self-reading without an inner loop is therefore not what EqR has over relax.

Round three: damping and inner loops (20k, preliminary). Table 3 and Figure 2. Lowering the merged-state retention from 0.40 ($\delta = 1$) to 0.18 ($\delta = 2$) gains seven points at 14k, and $\delta = 3$ ($r = 0.10$) is slightly worse. The twins keep H apart from L ($\cos = 0.25$–0.45) but are not better than relax $\delta = 1$ at 14k; the inner-loop reference is the best new arm at 14k. Whether an inner loop pays off may only show later in training, as EqR's own advantage grew between 20k and 50k; three arms are being extended to 50k.

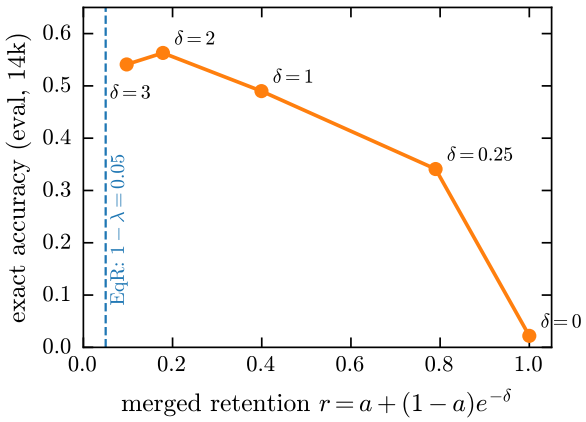


Figure 2: Relax, rate 3: accuracy at 14k against the retention r of the merged state (Prop. 5).

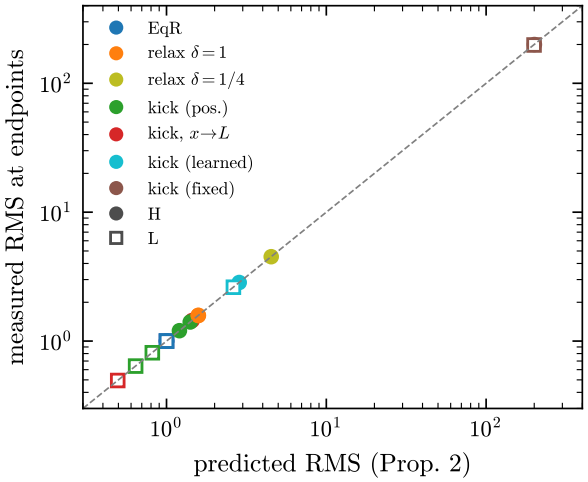


Figure 3: Endpoint state RMS against Prop. 2 (H filled, L open); all trained arms with a fixed point.

Table 3: Round three at 20k (W&B). Eval: whole test set, 16 iterations, at 14k. Train: training-batch exact accuracy averaged over 18–20k (depends on halting). $\cos(H, L)$ and mean ACT steps over the same window.

arm	eval@14k	train 18–20k	$\cos(H, L)$	ACT steps
EqR	62.7	72.0	0.41	7.9
relax, $\delta = 1$	49.0	68.4	1.00	8.2
kick, positive rates	53.5	64.8	0.58	8.8
relax, $\delta = \frac{1}{4}$	34.1	44.9	1.00	11.1
relax, $\delta = 2$	56.3	74.2	1.00	7.3
relax, $\delta = 3$	54.1	70.6	1.00	7.9
twin L, $\delta = 1$	50.4	62.4	0.45	9.3
twin L, $\delta = 2$	50.8	65.1	0.25	8.9
inner loop, self-read, $\delta = 1$	56.4	59.0	0.72	9.9

5 Test-time scaling

Figure 4 and Table 4. All working models improve steadily with depth. At 20k, relax ties EqR once compute is large (depth 1024: 90.4 vs. 91.4; depth 64 with 128 starts, most converged: 95.9 vs. 95.7). At 50k EqR leads everywhere. Relax's error decays like $D^{-0.39}$ without saturating, so it would need more than an order of magnitude more iterations to match EqR at 50k.

Positive kick at 20k is the outlier (Figure 5): at $\sigma = 0.5$ a single run is right only 37% of the time, yet some start is right for 99.1% of puzzles and a plurality vote over 128 starts reaches 96.8%, the best of the 20k models. Its relative noise is about 1.6 times EqR's (Corollary 1), which pushes it into a high-diversity regime; at noise 0.25 its single run is 69%. By 50k its L stream has grown (Prop. 2) and the regime is gone (any 93.8, vote 87.9).

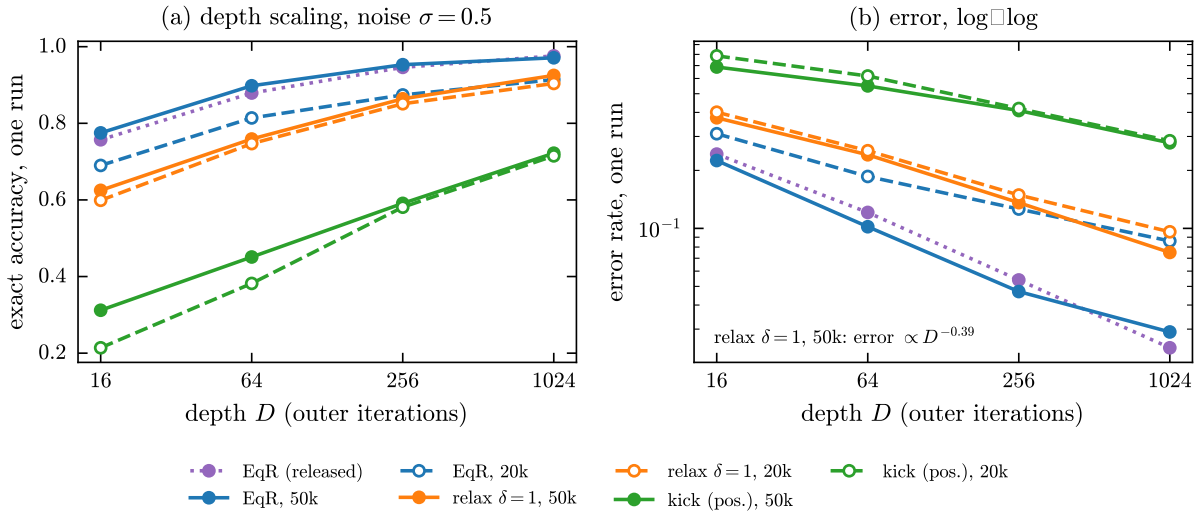


Figure 4: One-run accuracy against depth at $\sigma = 0.5$ (first 2048 test puzzles). Dashed/open: 20k steps; solid/filled: 50k; dotted: EqR's released checkpoint. (b) The error of relax falls like a power of depth; EqR's falls faster and then flattens near 3%.

Table 4: Test-time scaling at $\sigma = 0.5$, first 2048 test puzzles. Left: one start at depth D. Right: depth 64 with 128 starts, by selection rule (best in bold). The released EqR gives 98.6 top-4, as in EqR's README.

		one run, depth D				depth 64, 128 starts			
arm	steps	16	64	256	1024	converged	top-4	vote	any
EqR	20k	69.0	81.4	87.4	91.4	95.7	94.0	94.3	**95.9**
EqR	50k	77.5	89.8	95.3	97.1	98.7	98.0	98.0	**98.8**
EqR (released)	50k	75.7	87.9	94.6	97.6	**99.2**	98.6	98.7	**99.2**
relax, $\delta = 1$	20k	59.9	74.7	85.1	90.4	95.9	93.4	93.8	**96.0**
relax, $\delta = 1$	50k	62.5	75.9	86.4	92.5	96.6	95.3	95.6	**96.7**
kick, positive rates	20k	21.4	38.2	58.1	71.5	86.3	85.1	96.8	**99.1**
kick, positive rates	50k	31.2	45.1	59.1	72.2	93.7	88.5	87.9	**93.8**

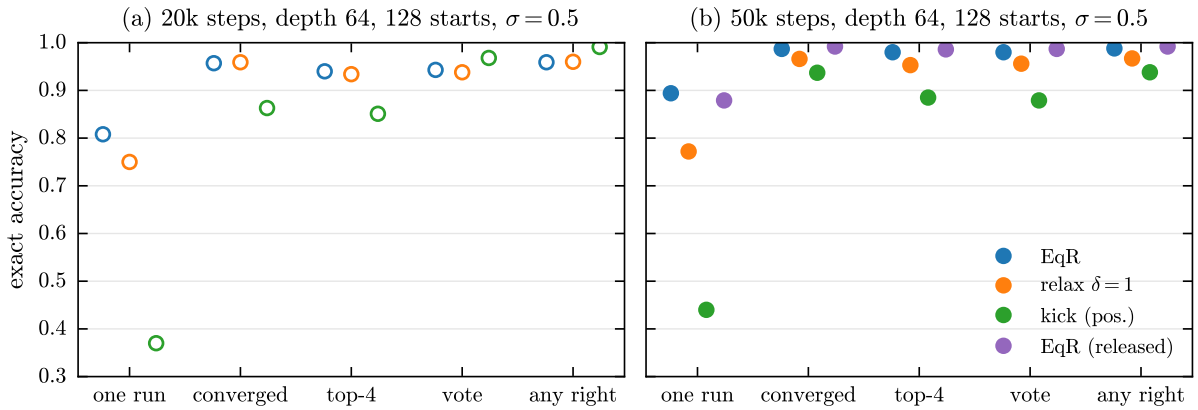


Figure 5: Depth 64, 128 starts, $\sigma = 0.5$: accuracy by selection rule. At 20k positive kick has the best vote and "any" but the worst single run and convergence selection; by 50k this has reversed.

6 Dynamics of the trained maps

All quantities in this section use the noise-free map unless stated, in float32, on the first 256 test puzzles with 16 random starts and 256 outer iterations, at the checkpoints' EMA weights.

6.1 Selection is solved; coverage is not

Table 5. For EqR and relax at $\sigma \leq 0.5$, the most converged start, the start with the largest halting logit and the verifier all agree with "any": whenever some start is right, each rule finds it, and the convergence score separates right from wrong runs with AUROC 0.98–1.00 (Figure 10). The exceptions are informative: relax $\delta = \frac{1}{4}$ at $\sigma = 0$ (AUROC 0.53; its right end states are not contracting, $\rho = 1.16$) and positive kick at 20k under $\sigma = 0.5$ (AUROC 0.86), the regime of its vote advantage. The differences between models are in *coverage* and in single-run accuracy, and these track path independence: at 50k EqR's starts reach 1.6 distinct answers per puzzle, relax's 2.6 and kick's 3.2 (Figure 6c).

6.2 Spectra at the end states

Observation 1 (Right answers attract, wrong end states do not)**.** In every working model the median spectral radius at right end states is below one and at wrong end states above one (Table 6, Figure 7). Wrong end states after 256 iterations are therefore not stable fixed points but slow transients or unstable sets, which is why convergence certifies correctness (§6.1) and why per-instance path independence separates right from wrong answers [2].

EqR's right answers sit close to criticality and move closer with training ($0.87 \to 0.93 \to 0.97$), as reported for converged Sudoku loopers [14] and in line with the drift of equilibrium models toward critical stability [3]. The coupled maps contract much harder (0.53–0.73). Leading eigenvalues at right answers are complex for EqR and relax at 20k (88%) but real at 50k (13%, 0%), and real for positive kick throughout (Figure 8): kick is *not* oscillatory at its answers, contrary to a momentum (underdamped) picture [10].

One caveat qualifies the comparison. For the coupled maps the spectral radius is almost the same at every right end state ($q_{10} \approx q_{90}$ in Table 6, e.g. $[0.53, 0.53]$ for relax at 20k), whereas EqR's varies from puzzle to puzzle. A puzzle-independent mode of the coupled update, rather than the puzzle's own constraint dynamics, may therefore set the coupled maps' ρ; the leading Ritz values (Figure 8) are real and tightly clustered, consistent with this. Identifying that mode is left open.

6.3 Escape under noise

Table 7 and Figure 9. Up to an arm-specific level $\sigma^\star$ right answers are kept with probability one while 15–33% of wrong end states are rescued; rescue peaks just below $\sigma^\star$. This is the selective destabilisation of spurious attractors known from Hopfield networks [1, 6], and it matches the published observation that noise lowers single-path quality but raises best-of-N [9]. $\sigma^\star$ is above 1 for EqR at 50k and relax at 20k, between 0.75 and 1 for EqR at 20k, the released EqR and kick at 20k, and between 0.5 and 0.75 for relax and kick at 50k. The best test noise for breadth is just below $\sigma^\star$, so it must be chosen per checkpoint: relax tolerated $\sigma = 0.75$ at 20k and collapses there at 50k (single run 3.7%), although its state scale is unchanged; its right answers attract less strongly at 50k (ρ $0.53 \to 0.68$). Relax $\delta = \frac{1}{4}$ rescues almost nothing: its wrong attractors are deep, which is why breadth does not help it.

Table 5: Multistart (256 puzzles, 16 starts, depth 256). "conv.": most converged; "q": largest halting logit; verifier: valid Sudoku keeping the givens. AUROC of EqR's convergence score for right vs. wrong runs. "answers": mean number of distinct final grids per puzzle.

arm	steps	single	any	vote	conv.	q	verifier	AUROC	answers	ΔPI
test noise $\sigma = 0$										
EqR	20k	73.5	83.2	79.3	83.2	83.2	83.2	0.984	4.5	0.038
EqR	50k	96.3	98.8	98.4	98.8	98.8	98.8	1.000	1.6	0.027
EqR (released)	50k	96.9	99.6	99.2	99.6	99.6	99.6	1.000	1.5	0.020
relax, $\delta = 1$	20k	64.1	70.7	68.8	70.7	70.7	70.7	1.000	5.2	0.043
relax, $\delta = 1$	50k	89.7	94.1	93.4	94.1	94.1	94.1	0.999	2.6	0.033
kick, positive rates	20k	63.4	70.3	66.8	70.3	70.3	70.3	0.998	4.0	0.036
kick, positive rates	50k	85.4	92.2	90.2	92.2	92.2	92.2	1.000	3.2	0.040
relax, $\delta = \frac{1}{4}$	20k	48.4	52.0	49.2	50.8	52.0	52.0	0.532	2.3	0.020
test noise $\sigma = 0.5$										
EqR	20k	89.4	97.3	96.9	97.3	97.3	97.3	0.999	2.7	0.050
EqR	50k	96.2	99.2	98.8	99.2	99.2	99.2	0.995	1.6	0.030
EqR (released)	50k	95.9	99.6	99.6	99.6	99.6	99.6	1.000	1.6	0.034
relax, $\delta = 1$	20k	82.6	94.1	91.4	94.1	94.1	94.1	0.997	3.7	0.075
relax, $\delta = 1$	50k	88.1	99.2	98.4	98.4	99.2	99.2	0.995	2.9	0.083
kick, positive rates	20k	63.0	97.7	91.8	85.2	71.5	97.7	0.864	6.8	0.236
kick, positive rates	50k	64.1	89.5	82.8	89.1	89.5	89.5	0.997	6.6	0.140
relax, $\delta = \frac{1}{4}$	20k	56.7	65.6	62.1	65.6	64.8	65.6	0.996	7.6	0.034

6.4 Inside one outer iteration

Figure 11. EqR at 20k shows HRM's *hierarchical convergence* [18]: within each recursion the L change falls roughly by half per update, then H moves and L restarts. The coupled maps have no such structure (Prop. 6); every update changes the state by a similar amount, and the second of two consecutive H updates at each recursion boundary changes almost nothing, so 2 of the 21 evaluations per outer iteration are nearly wasted by the alternating schedule.

6.5 EqR's inner loop goes through a flip bifurcation

Figures 12 and 13. At 20k the inner fixed point is stable (leading eigenvalues of DF near ± 0.6 for right puzzles; wrong puzzles sit at -1.00 to -1.03). At 50k a cluster of real eigenvalues sits at -0.93 to -1.07, and in the released checkpoint the leading eigenvalue is ≈ -1.9 to -2.0, i.e. $\mu \approx -2.1$ for the block. The inner fixed point is then unstable and z_L lives on a stable 2-cycle (two-step Jacobian eigenvalues 0.91–0.99). Because six inner updates are an even number, z_H always reads the same phase; in the released model z_H is then stationary to 4×10^{-7}.

The cycle carries no information the readout uses: odd inner-loop lengths, which make z_H read alternating phases, are as accurate as even ones (Table 8; ± 2.5 points at 128 puzzles). We read the period doubling as a by-product of strong negative feedback in the block combined with EqR's light damping, not as a computational mechanism. By Proposition 7 a twin coupling would hold such a cycle as a fixed point with $L \neq L'$; the twins' merge cosine is therefore a direct readout of whether the twin arms learn the same structure.

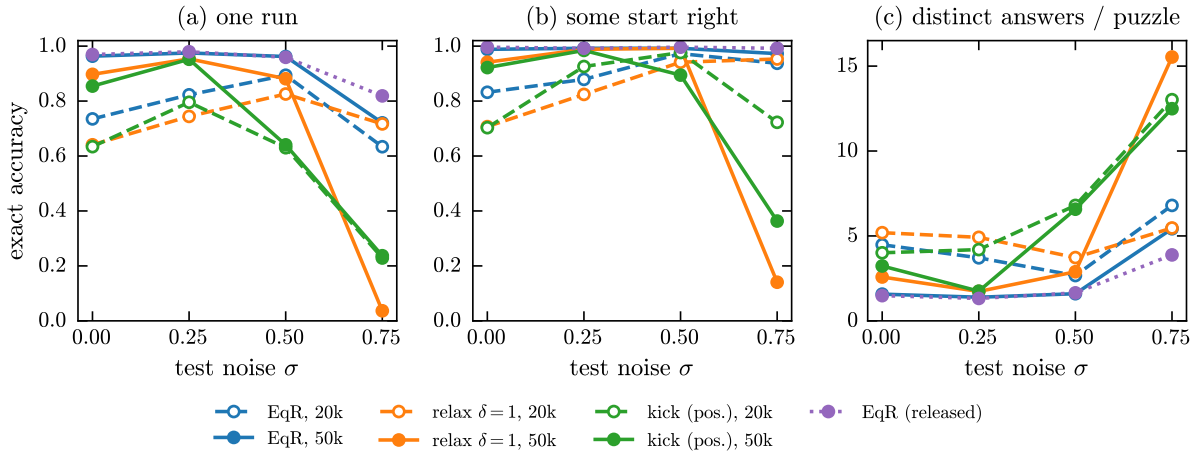


Figure 6: Multistart against test noise (256 puzzles, 16 starts, depth 256). (a) one run; (b) some start right; (c) distinct answers per puzzle. Relax at 50k collapses at $\sigma = 0.75$ although it tolerated it at 20k.

Table 6: Spectral radius of one outer iteration's Jacobian at noise-free end states (16 right and 16 wrong per arm, power iteration on JVPs), and the share of complex leading Ritz values (Arnoldi, 30 steps).

		right end states				wrong end states	
arm	steps	ρ	$[q_{10}, q_{90}]$	$\mathbb{C}$ top-1	$\mathbb{C}$ top-10	ρ	$\mathbb{C}$ top-1
EqR	20k	0.87	[0.81, 0.87]	88	85	1.52	25
EqR	50k	0.93	[0.90, 1.60]	12	71	4.76	44
EqR (released)	50k	0.97	[0.96, 0.99]	6	39	7.96	31
relax, $\delta = 1$	20k	0.53	[0.53, 0.53]	88	79	1.63	50
relax, $\delta = 1$	50k	0.67	[0.67, 0.67]	0	59	2.59	56
relax, $\delta = 1$, $x \to L$, self-read	20k	0.39	[0.39, 0.39]	0	40	2.41	38
relax, $\delta = \frac{1}{4}$	20k	1.16	[0.82, 2.35]	44	61	1.98	38
relax, $\delta = 0$	20k	1.00	[1.00, 1.00]	0	51	1.00	6
kick, positive rates	20k	0.73	[0.73, 0.74]	0	32	3.32	62
kick, positive rates	50k	0.66	[0.66, 0.66]	0	9	2.36	31
kick, learned rates	20k	1.26	[1.04, 1.72]	94	76	2.38	38
kick, fixed rates	20k	1.01	[0.99, 1.04]	0	46	0.99	0

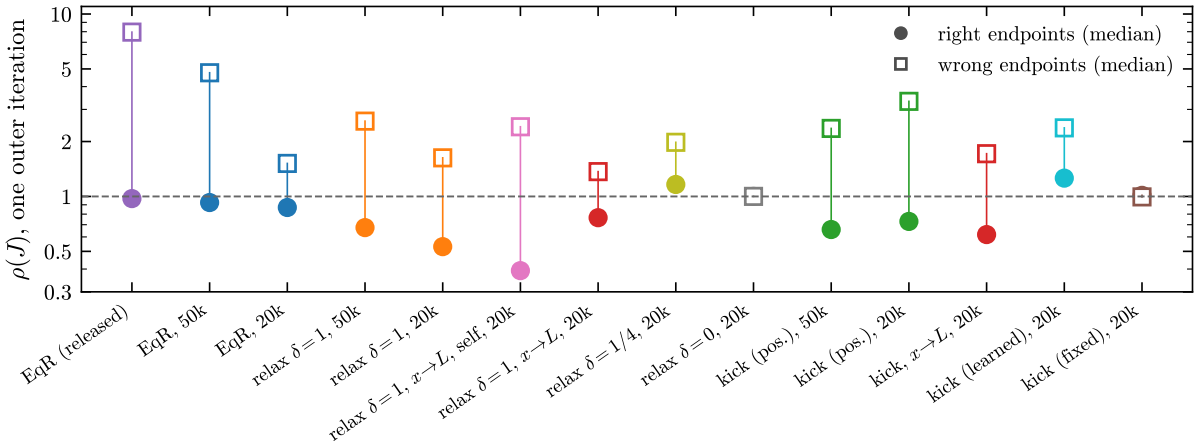


Figure 7: Median spectral radius of one outer iteration at right (filled) and wrong (open) end states. Dashed: $\rho = 1$.

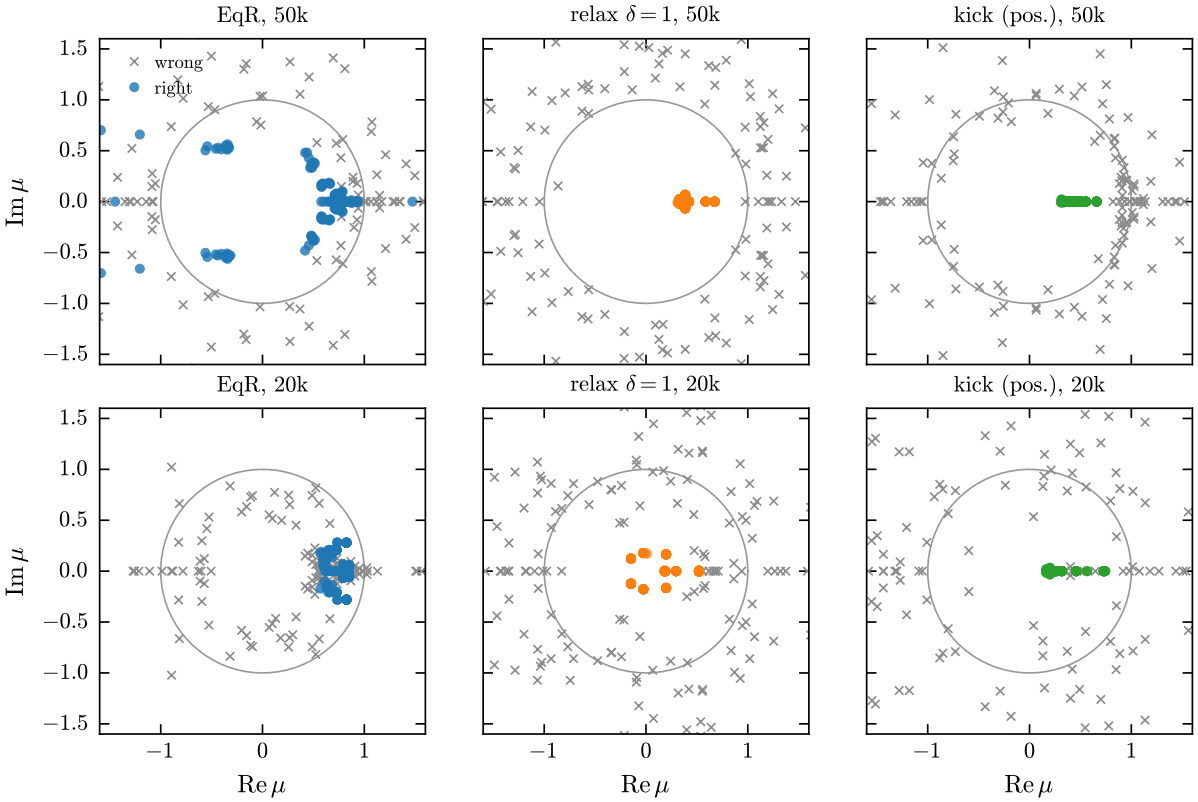


Figure 8: Leading Ritz values (top 10 per end state) of one outer iteration's Jacobian. Coloured: right end states; grey crosses: wrong end states; circle: $|\mu| = 1$.

Table 7: From noise-free end states, $K = 32$ iterations at noise σ: $P(\text{keep} \mid \text{right})$ / $P(\text{right} \mid \text{wrong})$.

arm	steps	$\sigma = 0.1$	0.25	0.5	0.75	1.0
EqR	20k	1.00 / 0.03	1.00 / 0.13	1.00 / 0.30	0.92 / 0.15	0.04 / 0.00
EqR	50k	1.00 / 0.07	1.00 / 0.16	1.00 / 0.27	1.00 / 0.04	0.98 / 0.03
EqR (released)	50k	1.00 / 0.06	1.00 / 0.14	1.00 / 0.23	0.99 / 0.11	0.08 / 0.00
relax, $\delta = 1$	20k	1.00 / 0.05	1.00 / 0.12	1.00 / 0.23	1.00 / 0.14	1.00 / 0.05
relax, $\delta = 1$	50k	1.00 / 0.07	1.00 / 0.24	1.00 / 0.15	0.04 / 0.00	0.02 / 0.00
kick, positive rates	20k	1.00 / 0.08	1.00 / 0.23	1.00 / 0.17	1.00 / 0.02	0.03 / 0.00
kick, positive rates	50k	1.00 / 0.07	1.00 / 0.33	1.00 / 0.17	0.33 / 0.00	0.01 / 0.00
relax, $\delta = \frac{1}{4}$	20k	1.00 / 0.02	1.00 / 0.06	1.00 / 0.07	1.00 / 0.05	1.00 / 0.04
relax, $\delta = 0$	20k	0.75 / 0.00	0.76 / 0.00	0.76 / 0.00	0.74 / 0.00	0.75 / 0.00
kick, fixed rates	20k	0.41 / 0.00	0.42 / 0.00	0.42 / 0.00	0.42 / 0.00	0.42 / 0.00

Table 8: Inner-loop length at test time (128 puzzles, one start, 32 outer iterations, local float32): exact accuracy at $\sigma = 0$ / $\sigma = 0.5$. Six is the trained value.

checkpoint	L=4	5	6	7	8
EqR 20k	75.0 / 81.2	75.0 / 80.5	71.1 / 79.7	73.4 / 79.7	72.7 / 78.1
EqR 50k	86.7 / 87.5	91.4 / 89.1	92.2 / 88.3	90.6 / 86.7	90.6 / 86.7
EqR released	90.6 / 89.1	94.5 / 85.9	94.5 / 84.4	93.0 / 88.3	94.5 / 85.2

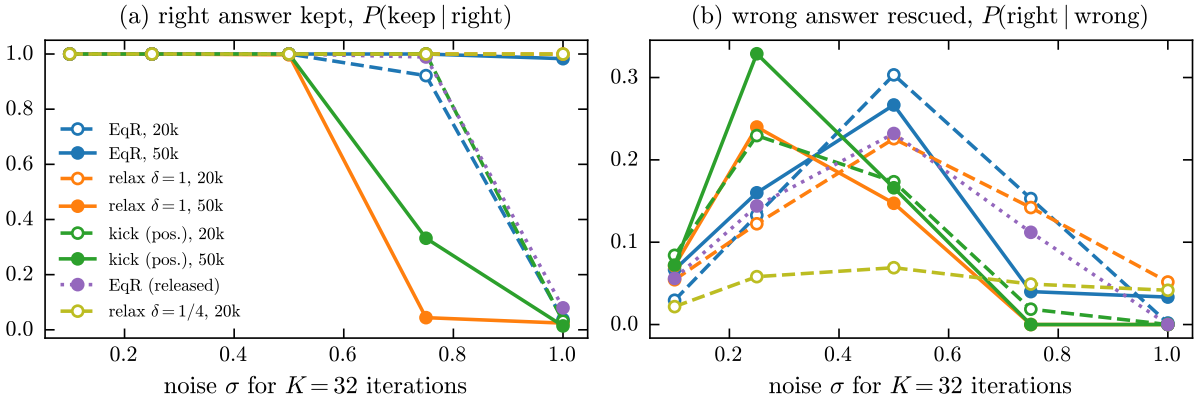


Figure 9: Escape from noise-free end states after $K = 32$ noisy iterations. (a) right answers kept; (b) wrong answers rescued.

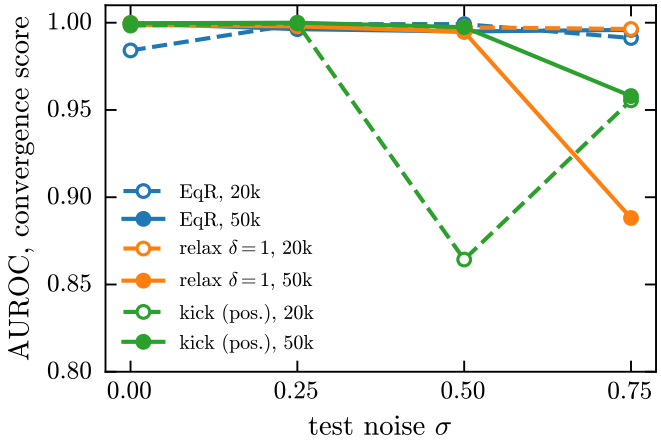


Figure 10: AUROC of EqR's convergence score for right vs. wrong runs, against test noise.

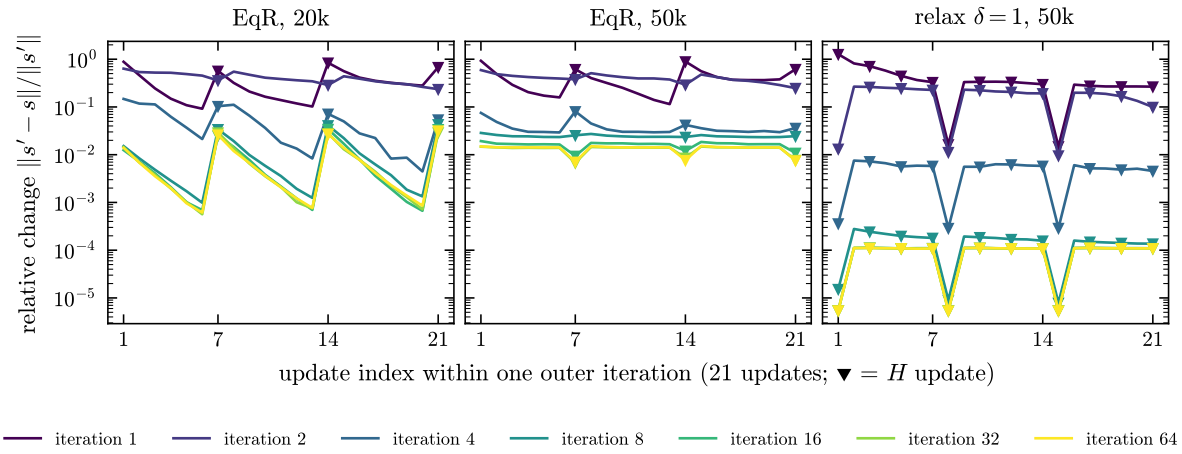


Figure 11: Median relative change per update inside one outer iteration (21 updates, ▼ = H updates), at outer iterations $1, \ldots, 64$ from a random start (64 puzzles). EqR at 20k shows hierarchical convergence: L settles within each recursion and is reset by H. At 50k the L changes no longer decay (the 2-cycle of §6.5). For relax the dips are the second of two consecutive H updates at recursion boundaries; the floor near 10^{-4} is TF32 round-off on the GPU.

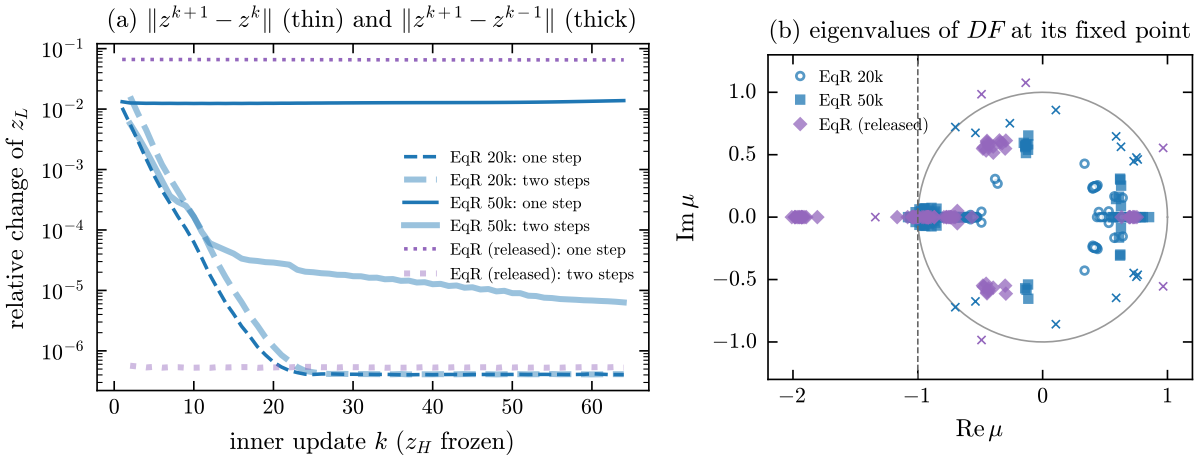


Figure 12: EqR's inner map $F(z_L) = 0.05 z_L + 0.95\mathcal{B}(z_L + z_H + x)$ with z_H frozen at an end state (16 puzzles, local float32). (a) Median one-step (thin) and two-step (thick) relative change over 64 inner updates: at 20k both vanish (fixed point); at 50k and in the released model the one-step change stays at 1% and 6% while the two-step change drops to round-off (exact 2-cycle). (b) Leading eigenvalues of DF at its fixed point (found by damped iteration); crosses: wrong puzzles.

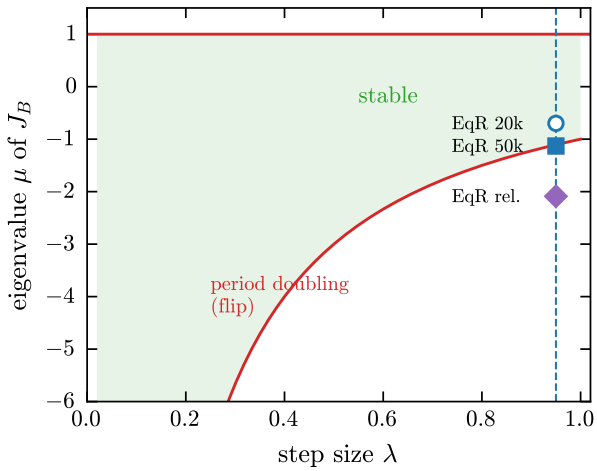


Figure 13: Stability of $F = (1-\lambda)z + \lambda\tilde{\mathcal{B}}(z)$ along a real eigen-direction μ of $D\tilde{\mathcal{B}}$ (Prop. 8). Markers: most negative leading eigenvalue of DF at the inner fixed point (median over right puzzles), mapped to μ at EqR's $\lambda = 0.95$.

7 Discussion

Which properties matter.

1. *Attracting right answers and non-attracting wrong ones.* Every working model has it, and it makes selection by convergence or halting essentially free. Models without it (relax $\delta = 0$, $\frac{1}{4}$; fixed and learned kick) also lose the certificate.
2. *Large, dominant basins for right answers (path independence).* This is where EqR leads, and it grows with training for every arm.
3. *A margin between the noise that breaks wrong attractors and the noise that breaks right ones.* It exists for all working models; its location $\sigma^\star$ is set by the state scale (exactly predictable, Prop. 2) and by contraction strength, and it moves with training.
4. *Damping of the common mode*, not the volume constraint. The undamped mode of relax is fatal; strong but not maximal damping is best.
5. *An inner equilibrium problem and near-critical contraction* distinguish EqR. The coupled maps contract harder, have an explicit inner equilibrium, and gain less from training. Whether restoring an implicit inner problem (twin, inner loop) closes the gap is the open question of round three.

Limitations. All runs use one seed; test-time numbers use 2048 puzzles (± 0.5 points) and the dynamical analysis 256 puzzles (± 1–2 points). Spectra use 16 right and 16 wrong end states per arm, and Ritz values from 30 Arnoldi steps approximate only the leading spectrum. GPU traces ran with TF32 matrix multiplies (round-off $\approx 10^{-4}$); the inner-loop analysis was redone locally in float32. The round-three results are from training curves only.

Next. Round three at 50k (relax $\delta = 2$, twin $\delta = 2$, inner loop), with depth×breadth evaluation and the same dynamical analysis; a per-checkpoint choice of test noise below $\sigma^\star$; a schedule without consecutive H updates; and seeds.

A Rates, scales and dimensionality

Table 9: Rates, the volume bound per outer iteration (Prop. 4), predicted vs. measured endpoint RMS of H / L (Prop. 2), and participation ratios $(\sum \ell_i)^2 / \sum \ell_i^2$ of the per-token state covariance.

arm	steps	γ	α	δ	vol. bound	RMS pred.	RMS meas.	PR H / L
EqR	20k	$1-\lambda = 0.05$ (self-read)			–	1.00 / 1.00	1.00 / 1.00	11.2 / 14.2
EqR	50k	$1-\lambda = 0.05$ (self-read)			–	1.00 / 1.00	1.00 / 1.00	10.5 / 14.4
EqR (released)	50k	$1-\lambda = 0.05$ (self-read)			–	1.00 / 1.00	1.00 / 1.00	10.1 / 12.5
relax, $\delta = 1$	20k	3.00	3.00	1.00	2.1×10^{-14}	1.58 / 1.58	1.58 / 1.58	14.3 / 14.3
relax, $\delta = 1$	50k	3.00	3.00	1.00	2.1×10^{-14}	1.58 / 1.58	1.58 / 1.58	12.8 / 12.8
relax, $\delta = \frac{1}{4}$	20k	3.00	3.00	0.25	2.1×10^{-14}	4.52 / 4.52	4.51 / 4.51	14.1 / 14.1
relax, $\delta = 0$	20k	3.00	3.00	–	2.1×10^{-14}	∞ / ∞	4028 / 4027	12.2 / 12.2
kick, positive rates	20k	0.94	1.24	–	8.5×10^{-6}	1.41 / 0.64	1.41 / 0.64	14.1 / 13.3
kick, positive rates	50k	0.80	1.77	–	6.5×10^{-7}	1.20 / 0.82	1.20 / 0.81	13.3 / 14.3
kick, learned rates	20k	0.32	0.43	–	1.7×10^{-2}	2.85 / 2.62	2.85 / 2.62	14.3 / 17.5
kick, fixed rates	20k	0.005	0.005	–	9.5×10^{-1}	201 / 200	200 / 198	28.5 / 36.9
kick, positive, $x \to L$ only	20k	1.11	1.18	–	6.0×10^{-6}	1.45 / 0.50	1.44 / 0.49	14.0 / 13.6

B Reproduction

- Training and evaluation: `jobs/looped/sudoku_eqr.slurm`, `jobs/looped/sudoku_scaling.slurm`; summary tables: `looped/eqr/summarize_sudoku.py`.
- Dynamical analysis: `looped/eqr/export_for_analysis.py`, `looped/eqr/analyze_maps.py` (GPU job `jobs/looped/sudoku_maps.slurm`), `looped/eqr/summarize_maps.py`.
- This note: `looped/eqr/paper/make_figures.py` and `make_tables.py`, from the data in `looped/eqr/paper/data/`.

References

[1] Daniel J. Amit, Hanoch Gutfreund, and H. Sompolinsky. Storing infinite numbers of patterns in a spin-glass model of neural networks. *Physical Review Letters*, 55(14):1530–1533, 1985.

[2] Cem Anil, Ashwini Pokle, Kaiqu Liang, Johannes Treutlein, Yuhuai Wu, Shaojie Bai, J. Zico Kolter, and Roger Grosse. Path independent equilibrium models can better exploit test-time computation. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2211.09961.

[3] Shaojie Bai, Vladlen Koltun, and J. Zico Kolter. Stabilizing equilibrium models by Jacobian regularization. In *International Conference on Machine Learning*, 2021. arXiv:2106.14342.

[4] Vivek S. Borkar. Stochastic approximation with two time scales. *Systems & Control Letters*, 29(5):291–294, 1997.

[5] Jehoshua Bruck. On the convergence properties of the Hopfield model. *Proceedings of the IEEE*, 78(10):1579–1585, 1990.

[6] A. C. C. Coolen. Statistical mechanics of recurrent neural networks I. Statics. *arXiv preprint cond-mat/0006010*, 2000.

[7] Guilherme França, Jeremias Sulam, Daniel P. Robinson, and René Vidal. Conformal symplectic and relativistic optimization. *Journal of Statistical Mechanics: Theory and Experiment*, 2020. arXiv:1903.04100.

[8] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. *arXiv preprint arXiv:2502.05171*, 2025.

[9] Alexi Gladstone, Ganesh Nanduru, Md Mofijul Islam, Peixuan Han, Hyeonjeong Ha, Aman Chadha, Yilun Du, Heng Ji, Jundong Li, and Tariq Iqbal. Energy-based transformers are scalable learners and thinkers. *arXiv preprint arXiv:2507.02092*, 2025.

[10] Gabriel Goh. Why momentum really works. *Distill*, 2017. https://distill.pub/2017/momentum/.

[11] Eric Goles and Jorge Olivos. Comportement périodique des fonctions à seuil binaires et applications. *Discrete Applied Mathematics*, 3(2):93–105, 1981.

[12] Benhao Huang, Zhengyang Geng, and Zico Kolter. Equilibrium reasoners: Learning attractors enables scalable reasoning. *arXiv preprint arXiv:2605.21488*, 2026.

[13] Alexia Jolicoeur-Martineau. Less is more: Recursive reasoning with tiny networks. *arXiv preprint arXiv:2510.04871*, 2025.

[14] Asher Labovich. Stability and generalization in looped transformers. *arXiv preprint arXiv:2604.15259*, 2026.

[15] Robert McLachlan and Matthew Perlmutter. Conformal Hamiltonian systems. *Journal of Geometry and Physics*, 39(4):276–300, 2001.

[16] Sajad Movahedi, Vera Milovanović, Shlomo Libo Feigin, Alexander Theus, Thomas Hofmann, Valentina Boeva, T. Konstantin Rusch, and Antonio Orvieto. Fixed-point reasoners: Stable and adaptive deep looped transformers. *arXiv preprint arXiv:2606.18206*, 2026.

[17] Michael E. Sander, Pierre Ablin, Mathieu Blondel, and Gabriel Peyré. Momentum residual neural networks. In *International Conference on Machine Learning*, 2021. arXiv:2102.07870.

[18] Guan Wang, Jin Li, Yuhao Sun, Xing Chen, Changling Liu, Yue Wu, Meng Lu, Sen Song, and Yasin Abbasi Yadkori. Hierarchical reasoning model. *arXiv preprint arXiv:2506.21734*, 2025.